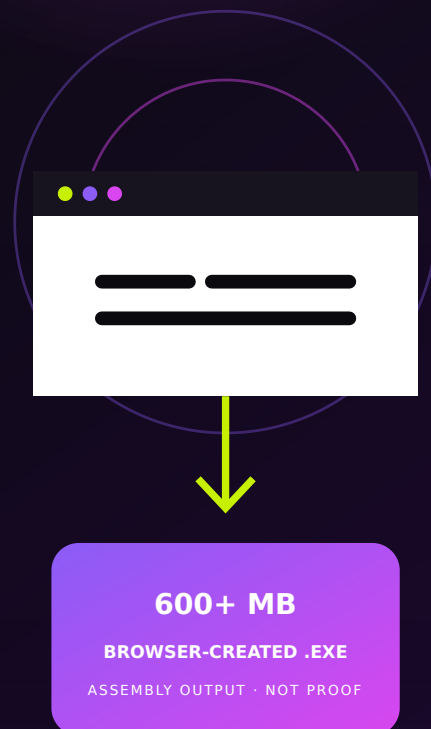


SOURTRADE BROWSER- BUILT EXECUTABLE

Hunting for oversized Windows executables created by browsers after in-page Bun assembly.

HUNTING**CONFIDENCE: MEDIUM****REPORT DATE****28 July 2026****VERSION****v1.0****PRIMARY TELEMETRY****Defender XDR ·
DeviceFileEvents****ATT&CK****T1204.002 · Malicious
File**

02

EXECUTIVE ASSESSMENT

What is confirmed, what is inferred, and what the hunt can establish.

7.8 / 10**01**

OPERATIONAL VALUE

Medium**02**

ANALYTICAL CONFIDENCE

Hunting**03**

OPERATIONAL READINESS

SOURCE-CONFIRMED FACTS

- SourTrade registers a ServiceWorker and builds a SharedWorker from embedded JavaScript.
- The worker retrieves build instructions and a clean Bun runtime.
- The published response specifies a 665,136,399-byte pseudorandom stream.
- The browser returns the assembled executable as a same-origin attachment.

ANALYTICAL ASSESSMENT

A browser-created executable above 600 MB is a practical campaign-oriented endpoint pivot. The size condition is supported by one published sample, so it must be baselined before promotion.

Evidence boundary

A match does not prove in-browser assembly, malicious bytecode, execution or credential theft.

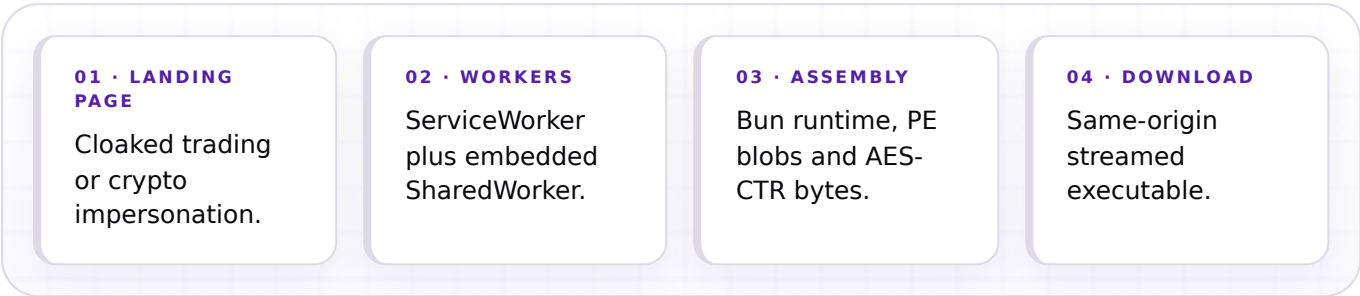
Detection hypothesis

If an attacker uses SourTrade's documented browser assembly flow, endpoint file telemetry should show a browser creating a very large executable. The event may be distinguished from legitimate downloads through file size, origin metadata, PE inspection, user intent and follow-on execution.

03

THREAT MECHANICS

Source-observed browser-side assembly and the endpoint-visible boundary.



STAGE	SOURCE EVIDENCE	ENDPOINT OBSERVABLE	DETECTION VALUE
Fingerprinting	Victim/analyst separation	Not reliably visible in endpoint file events	None for this hunt
Instructions	/config returns template, seed, size and runtime URL	Payload body unavailable in DeviceFileEvents	Network concept only
Assembly	Browser combines clean runtime, blobs and generated bytes	In-memory operations not exposed by selected table	Evidence gap
File creation	ServiceWorker returns final executable as attachment	Browser process, file path, size, hash and origin fields	Primary hunt signal

Unconfirmed elements

The public report does not demonstrate that every SourTrade build exceeds 600 MB. It does not provide a durable filename, stable hash or invariant landing domain.



TELEMETRY AND EVIDENCE MODEL

Required schema, optional enrichment and known blind spots.

Microsoft Defender XDR - DeviceFileEvents

FIELD	RELIABILITY	PURPOSE	LEVEL
ActionType	Documented; value portal-dependent	Select file creation events	Required
InitiatingProcessFileName	Documented	Identify supported browser writers	Required
FileName, FolderPath	Documented	Executable and destination	Required
FileSize	Documented	Source-supported size pivot	Required
FileOriginUrl, FileOriginReferrerUrl	Documented; population environment-dependent	MotW and origin review	Optional
SHA1	Documented; may be empty	Sandbox and prevalence pivot	Enrichment

DATA QUALITY REQUIREMENTS

- Microsoft Defender for Endpoint deployed.
- Browser file creation captured on Windows endpoints.
- FileSize populated for large downloads.
- Portal confirms FileCreated semantics.

UNAVAILABLE IN THIS TABLE

- ServiceWorker or SharedWorker registration.
- /config response body and template.
- PE section names and JavaScriptCore bytecode.
- Proof that the file was executed.

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PRIMARY HUNTING LOGIC

Direct, single-table KQL designed for evidence collection.

KQL · HUNTING

```
// Hunting query - Untested implementation sketch
DeviceFileEvents
| where ActionType == "FileCreated"
| where InitiatingProcessFileName in~ (
    "chrome.exe", "msedge.exe", "firefox.exe",
    "brave.exe", "opera.exe")
| where FileName endswith ".exe"
| where FileSize >= 600000000
| project Timestamp, DeviceId, DeviceName,
    InitiatingProcessAccountUpn, InitiatingProcessFileName,
    FileName, FolderPath, FileSize, SHA1,
    FileOriginUrl, FileOriginReferrerUrl, FileOriginIP, ReportId
| order by Timestamp desc
```

CONDITION	EVIDENCE	RATIONALE
Browser writer	Source: final stream returned through browser	Direct endpoint representation of delivery
.exe	Source: assembled Windows PE executable	Excludes non-executable downloads
≥ 600,000,000 bytes	Published random component: 665,136,399 bytes	Campaign-oriented margin below source sample

What a match means

A browser created a Windows executable at least 600 MB in size.

What it does not prove

Assembly method, payload content, execution, theft, campaign identity or malicious intent.

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TRIAGE AND INVESTIGATION

Turn the size-based match into defensible evidence.

1 · ESTABLISH USER INTENT

Confirm whether the user intentionally downloaded a large developer runtime, game, virtual appliance or installer.

2 · INSPECT ORIGIN

Review FileOriginUrl, referrer, MotW Zone.Identifier and browser history. Compare registered domain and impersonated brand.

3 · INSPECT THE PE

Verify signer, original filename, section table, presence and size of .bun, embedded JavaScriptCore bytecode and inflated regions.

4 · PIVOT TO EXECUTION

Search DeviceProcessEvents for the hash/path. Review child processes, credential access, persistence and outbound connections.

OUTCOME	EVIDENCE	ACTION
Benign	Expected source, approved software, trusted signature and known user intent	Document and tune narrowly
Suspicious	Brand mismatch, unknown signer, unusual .bun section or absent business purpose	Contain pending analysis
Likely malicious	Source-aligned PE structure plus execution or malicious follow-on behavior	Isolate, acquire artifacts, reset exposed credentials
Insufficient evidence	Missing file bytes, origin fields or sensor coverage	Preserve event and close as unresolved

07

FALSE POSITIVES AND VALIDATION

Known overlap, controlled tests and promotion criteria.

BENIGN SCENARIO	LIKELIHOOD	IMPACT	TUNING ACTION
Large game or creative-suite installer	Medium	High analyst cost	Allowlist verified publisher + origin + path, never filename alone
Developer downloads Bun or bundled runtime	Low-Medium	Moderate	Validate team, source domain and signature
Virtual appliance or offline bundle	Low	Moderate	Document software-distribution workflow
Missing or delayed file-size telemetry	Environment-dependent	False negative	Run controlled browser downloads and measure coverage

VALIDATION STEPS

1. Download benign executables above and below 600 MB with each supported browser.
2. Confirm FileCreated, writer, size and origin fields.
3. Run the hunt over 30-90 days and label legitimate classes.
4. Inspect a controlled Bun standalone executable to validate PE enrichment.

DECISION CRITERIA

Success: events retain correct browser writer and file size, with manageable legitimate volume.

Rejection: common business downloads dominate or FileSize is unreliable.

Failure: MDE coverage is absent or file creation is not captured.

Promotion gate

Do not promote until repeated malicious samples confirm a durable invariant beyond one file-size observation, or local telemetry demonstrates a specific combination of origin, PE metadata and follow-on execution with controlled overlap.



ATT&CK, CONCLUSION AND REFERENCES

Coverage statement, residual risk and source control.

TECHNIQUE	NAME	SUPPORT	COVERAGE
T1204.002	User Execution: Malicious File	The source documents a click-driven executable download intended for user execution.	Partial: file creation only; execution is not required by the query.

ANALYST CONCLUSION

The published browser-assembly mechanism is technically strong research, but endpoint telemetry cannot directly observe its key in-memory stages. The 600 MB browser-created executable is a useful campaign-shaped hunt with Medium confidence, not a durable production rule.

RESIDUAL RISK

- Smaller future builds bypass the hunt.
- Other browsers are excluded until validated.
- Execution and post-compromise behavior require separate pivots.
- Origin fields may be absent.

REFERENCES

1. Confiant Threat Intelligence, “SourTrade: Browser-Assembled Malware Delivered Through Malvertising,” 23 Jul 2026. [https://blog\[.\]confiant\[.\]com/p/sourtrade-browser-assembled-malware](https://blog[.]confiant[.]com/p/sourtrade-browser-assembled-malware)
2. Microsoft Learn, “DeviceFileEvents table in the advanced hunting schema,” updated 22 Jun 2026. [https://learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-devicefileevents-table](https://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-devicefileevents-table)
3. MITRE ATT&CK, T1204.002. [https://attack\[.\]mitre\[.\]org/techniques/T1204/002/](https://attack[.]mitre[.]org/techniques/T1204/002/)

Document control

Version 1.0 · 28 Jul 2026 · Status: Hunting · Confidence: Medium · Owner: Detection Engineering · No customer data or private indicators.